ANGELINA LESNIAK

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Personal Website | Google Scholar | LinkedIn

EDUCATION

University of Wisconsin - Madison *Fall 2025 – Fall 2027*
MS in Electrical Engineering GPA: 3.917
Graduate Certificate in Energy Analysis and Policy
Milwaukee School of Engineering *Fall 2021 – Spring 2025*
BS in Electrical Engineering GPA: 3.890

RESEARCH FOCUS

Grid-aware and carbon-aware modeling of flexible energy resources, energy storage, and industrial loads.

RESEARCH EXPERIENCE

GRADUATE RESEARCH ASSISTANT *05/2026 - 08-2026*
POST-BACHELOR'S RESEARCH ASSISTANT *05/2025 - 08-2025*
Center for Non-Linear Studies (CNLS), Los Alamos National Laboratory Los Alamos, NM

- Built an end-to-end multi-stage power-system simulation model combining unit commitment, DC optimal power flow, carbon and price signals, and load-shifting optimization on the RTS-GMLC test system.
- Evaluated carbon and price-based load-shifting signals across short and long-term simulations, identifying how alignment between grid cost and emissions merit orders affects signal performance and carbon-aware flexible-load operation.
- Developed a novel analytical method to compute multi-dimensional validity regions of Locational Marginal Carbon Emissions (LMCEs) under DC optimal power flow.
- Developed an iterative extension method that recovers larger LMCE-invariant load-shifting regions while maintaining prediction accuracy across multiple test networks.

RESEARCH ASSISTANT *05/2024 - Present*
WISPO Laboratory Group, University of Wisconsin - Madison Madison, WI

- Conducted research on the scheduling of hybrid renewable (wind/solar farm) and hydrogen electrolyzer systems under the supervision of Dr. Line Roald at the University of Wisconsin-Madison.
- Analyzed day-ahead energy market data and developed models to optimize the efficiency of modular hydrogen electrolyzer systems and their interactions with the electric grid.
- Presented research findings at Los Alamos National Laboratory and IEEE-GM poster sessions, as well as with an academic paper.

PROFESSIONAL EXPERIENCE

ELECTRICAL ENGINEERING INTERN *02/2022 – 05/2025*
KW Mission Critical Engineering Milwaukee, WI

- Supported the design of electrical power systems for large-scale, mission-critical data centers, working across multiple phases of engineering design and construction.
- Independently produced complete electrical drawing packages for mission-critical data center projects, coordinating designs across disciplines and successfully advancing drawings through AHJ permit review and approval.
- Performed electrical calculations, system modeling, equipment selection, and design documentation to support reliable and code-compliant power distribution systems.

- Coordinated electrical designs with mechanical, architectural, civil, and fire-protection teams to resolve multidisciplinary design requirements and conflicts.
- Traveled to multiple data center construction site locations to analyze and provide real time solutions to complex problems on-site.

ELECTRICAL ENGINEERING INTERN *12/2022 – 08/2023*
AHEAD Engineered Solutions Libertyville, IL

- Participated in a team of software, mechanical, and electrical engineers to design integrated hardware and software solutions.
- Collaborated on the development of the Kori cart, an industry-first modular mobile medical cart with integrated artificial intelligence designed to improve patient care in hospitals.
- Designed and conducted tests on prototypes to improve functionality.
- Prepared comprehensive reports based on test results to advise clients.
- Worked with a team to organize and provide accurate information for product lifecycle management.

AWARDS & GRANTS

- LANL Center for Non-Linear Studies Graduate Fellowship 2026
- Grainger Power Engineering Award, University of Wisconsin-Madison 10/2025
- LANL Center for Non-Linear Studies Post-Baccalaureate Fellowship 2025
- NSF REU fellowship, University of Wisconsin-Madison 2024
- MSOE Dean's and Honors Lists 2021 - 2024
- MSOE Academic Merit Scholarship 2021 - 2024

PUBLICATIONS & PRESENTATIONS

MANUSCRIPTS

- **Angelina Lesniak,** Noah Rhodes, and Line Roald. "Carbon-Aware Load Shifting Across Multiple Buses: Increasing Accuracy and Impact through Validity-Constrained Shifting." 60th Hawaii International Conference on System Sciences (HICSS-60), 2027**. Accepted for publication.**
- **Angelina Lesniak,** Andrea Gloppen Johnsen, Noah Rhodes, and Line Roald. "Advanced Scheduling of Electrolyzer Modules for Grid Flexibility." 2025 IEEE Power & Energy Society General Meeting (PESGM), 2025. DOI: 10.1109/PESGM52009.2025.11225446.

INVITED TALKS & ORAL PRESENTATIONS

- **Angelina Lesniak,** Noah Rhodes, and Line Roald. "Carbon-Aware Load Shifting Across Multiple Buses: Increasing Accuracy and Impact through Validity-Constrained Shifting." 2026 INFORMS Annual Meeting, San Francisco, CA, November 2026.

POSTER PRESENTATIONS

- **Angelina Lesniak,** Noah Rhodes, and Line Roald. "Carbon-Aware Load Shifting Across Multiple Buses: Increasing Accuracy and Impact through Validity-Constrained Shifting." PowerUp 2026 Conference, Boulder, CO, September 2026.
- **Angelina Lesniak,** Noah Rhodes, and Line Roald. "Carbon-Aware Load Shifting Across Multiple Buses: Increasing Accuracy and Impact through Validity-Constrained Shifting." Los Alamos National Laboratory Student Symposium, Los Alamos, NM, August 2026.

- **Angelina Lesniak,** Noah Rhodes, Andrea Gloppen-Johnsen, and Line Roald. “Advanced Scheduling of Electrolyzer Modules for Grid Flexibility.” IEEE Power & Energy Society General Meeting (PESGM), Austin, TX, July 2025.
- **Angelina Lesniak,** Noah Rhodes, Andrea Gloppen-Johnsen, and Line Roald. “Advanced Scheduling of Electrolyzer Modules for Grid Flexibility.” Los Alamos Grid Science Winter School and Conference, Santa Fe, NM, January 2025.

PROFESSIONAL MEMBERSHIPS

- Tau Beta Pi Engineering Honor Society
- Society of Women Engineers (SWE)

LEADERSHIP AND SERVICE

- Community Engagement for the University of Wisconsin-Madison Upward Bound Workshops
- Executive Board Member, Tau Beta Pi, Wisconsin Delta chapter
- Founding Executive Board Member, MSOE Mug Club
- Ice Age Trail Alliance Volunteer